# V4.2 Toy Model 4I
# Tensorial Relational Action and Metric-Interface Derivation

*Rev A - Exploratory Gravity / Metric Bridge Report*

*Date: 2026.05.26* | *Continuation after Toy Model 4H*

**Status: Toy Model 4I upgrades the scalar response field Phi into a tensor response field hbar_mu_nu. It derives tensor Poisson equations from a tensorial action and tests whether the trace-reversed metric interface is selected. It does not derive the full nonlinear Einstein-Hilbert action, physical gravity, dark matter, dark energy, or physical 3T.**

## 1. Purpose

Toy Model 4H derived the scalar graph response law from a native action, but still used a trace-reversed metric interface as an ansatz. Toy Model 4I upgrades the response variable itself from a scalar Phi to a tensorial field hbar_mu_nu.

The narrow question is not whether full General Relativity has been derived. The question is whether a tensorial graph action can produce component-wise metric-response equations and whether the trace-reversed interface is mathematically selected by the balance test.

## 2. Critical result before details

**4I is a partial success, not a full derivation. The tensorial action produces the component-wise response equations. The trace-reversed interface is strongly selected by the balance test: the trace-reversed map works, while the direct h=hbar map fails. However, this is selection by consistency, not a first-principles derivation of the full spin-2 gauge action.**

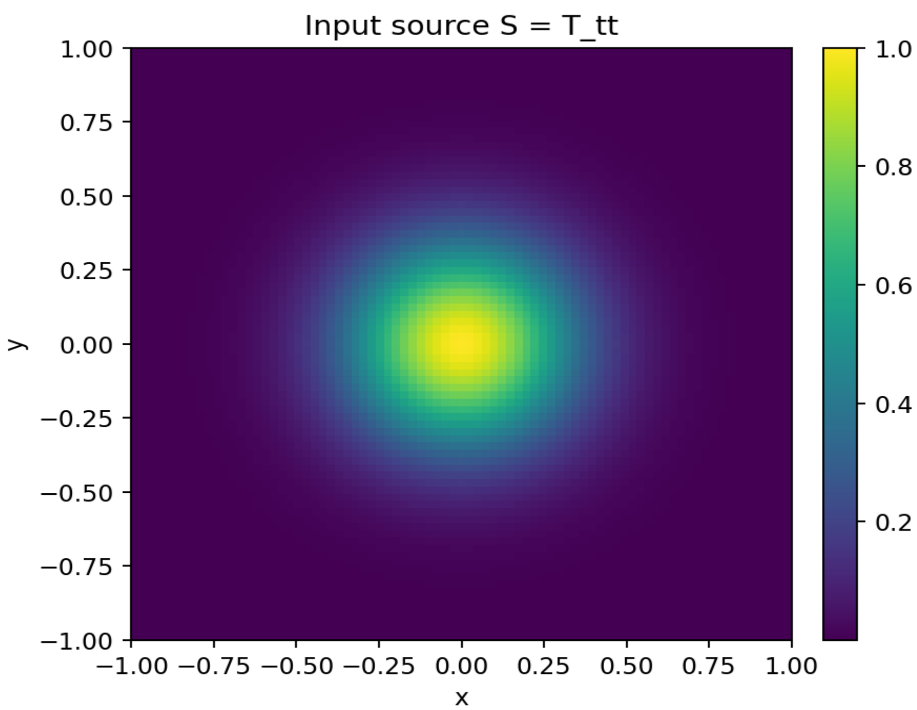
## 3. Tensorial relational action

The scalar field Phi is replaced by a symmetric tensor response field hbar_mu_nu. In this simplified toy model, each tensor component carries a smoothness cost and couples to the corresponding source component:

```
A[hbar] =
integral [ 1/2 sum_mu_nu |grad hbar_mu_nu|^2
           - sum_mu_nu T_mu_nu hbar_mu_nu ] dA
```

Meaning of the terms:

- hbar_mu_nu is the tensor response field.
- T_mu_nu is the relational stress/source tensor.
- The gradient term penalizes uneven tensor response between neighboring nodes.
- The coupling term rewards response where relational source exists.
- This is a linear weak-field tensor action, not the full Einstein-Hilbert action.

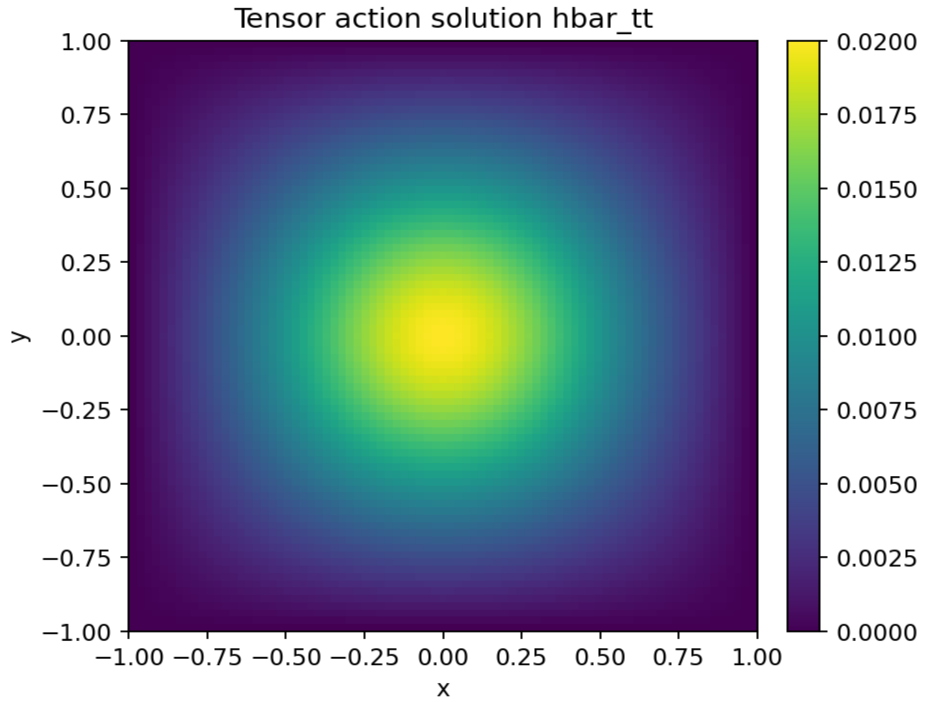


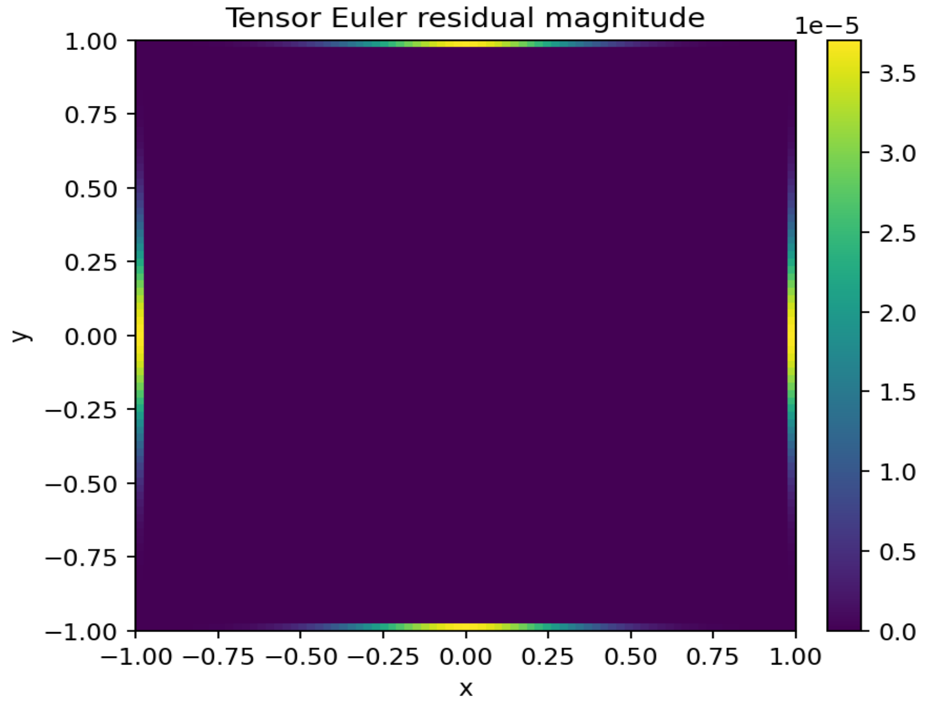
## 4. Tensorial variation

Varying the action with respect to hbar_mu_nu gives one Poisson-type equation per tensor component:

```
delta A / delta hbar_mu_nu = 0

=> -nabla^2 hbar_mu_nu - T_mu_nu = 0

=> nabla^2 hbar_mu_nu = -T_mu_nu
```

This is the tensorial generalization of the scalar result from 4H.




| Tensor-action diagnostic | Value |
|---|---|
| Action at zero tensor field | 0.000000e+00 |
| Action at tensor stationary field | -6.662312e-04 |
| Dust tensor Euler residual max | 8.046341e-14 |
| Dust tensor Euler residual RMS | 3.071363e-15 |
| Anisotropic tensor Euler residual max | 8.046341e-14 |
| Anisotropic tensor Euler residual RMS | 3.135906e-15 |

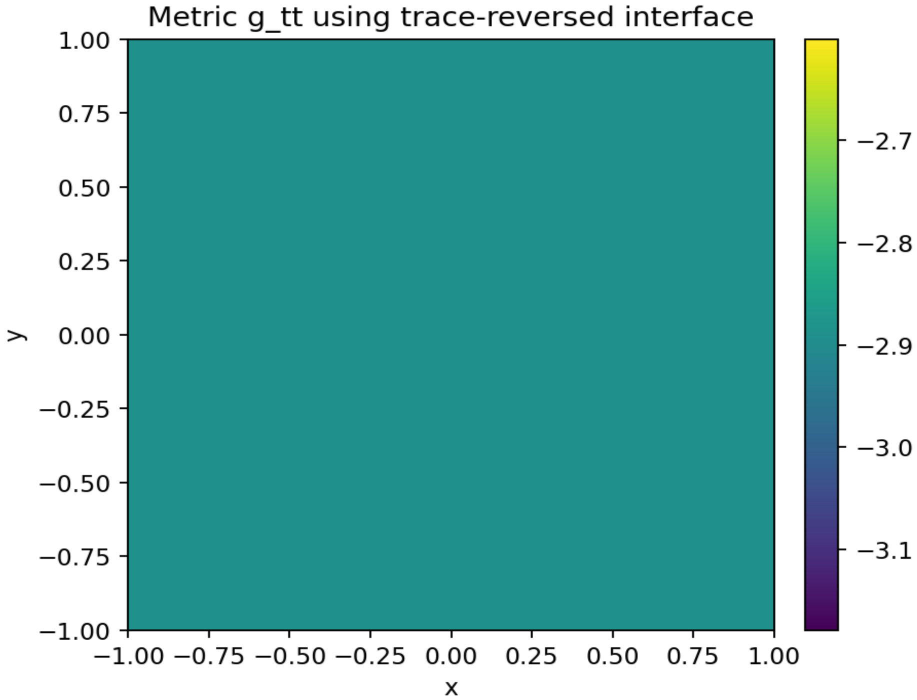
**The Euler residual is at numerical precision. The tensor action solves its own component-wise equilibrium equations.**

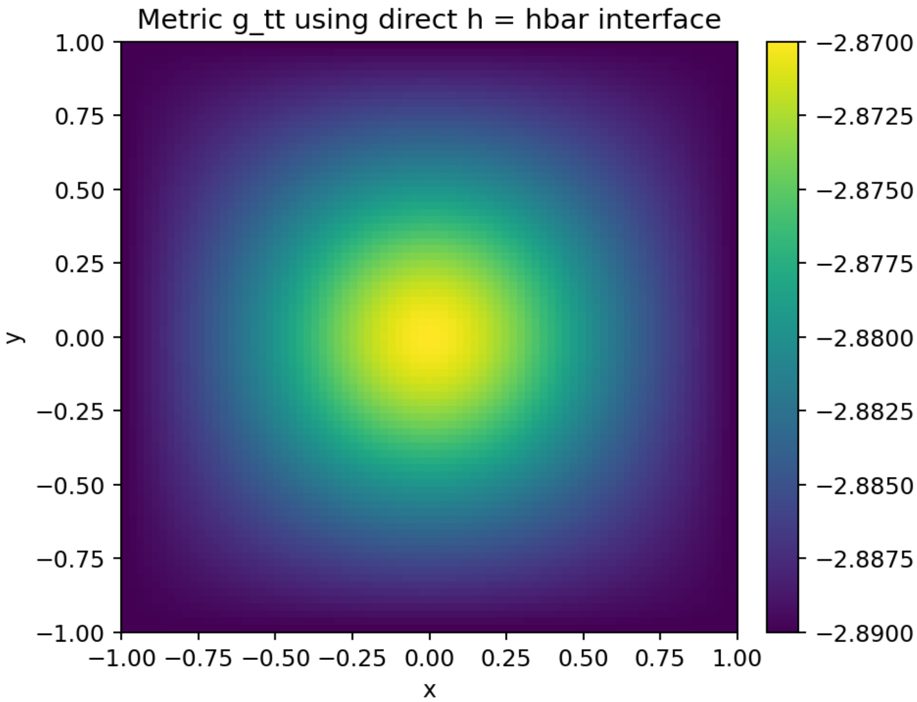
## 5. Metric-interface test: trace-reversed versus direct

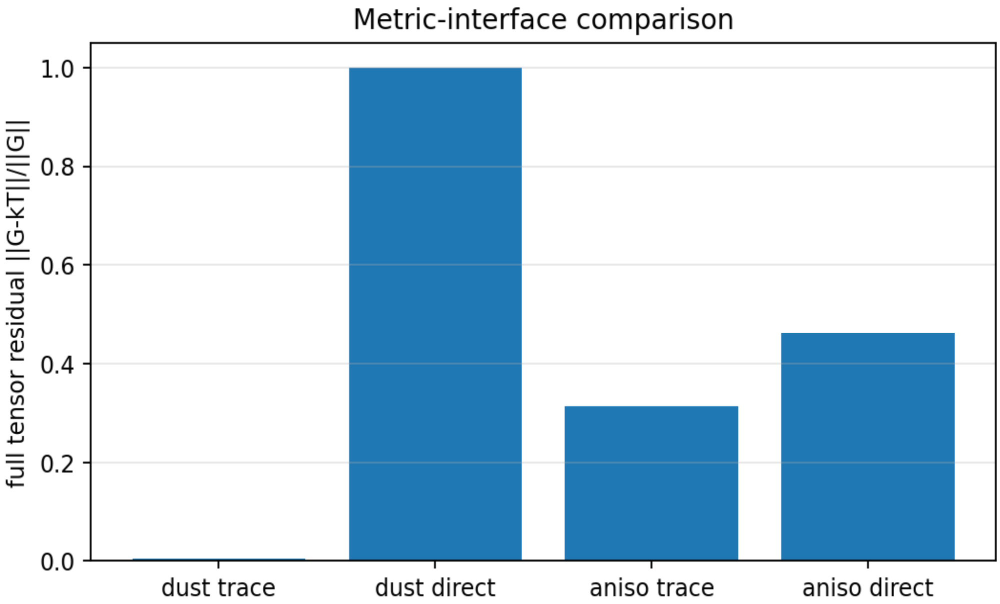
Two ways of converting the tensor response into a metric perturbation were tested.

```
Direct interface:
h_mu_nu = hbar_mu_nu

Trace-reversed interface:
h_mu_nu = hbar_mu_nu - eta_mu_nu hbar_trace/(D-2)

Metric:
g_mu_nu = eta_mu_nu + h_mu_nu
```

If the trace-reversed interface were merely decorative, both maps would perform similarly. They do not.



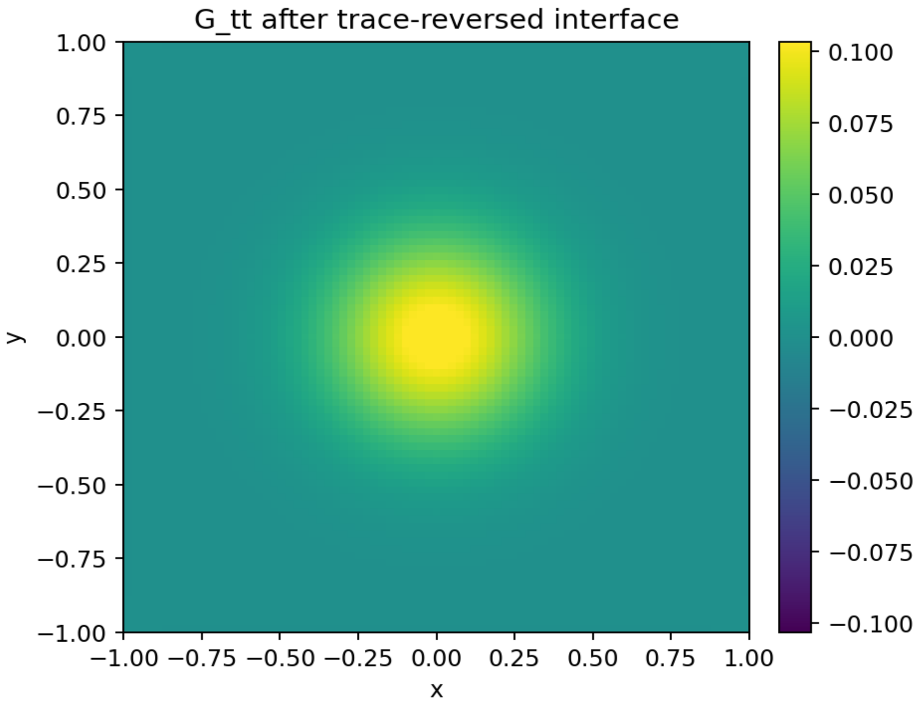



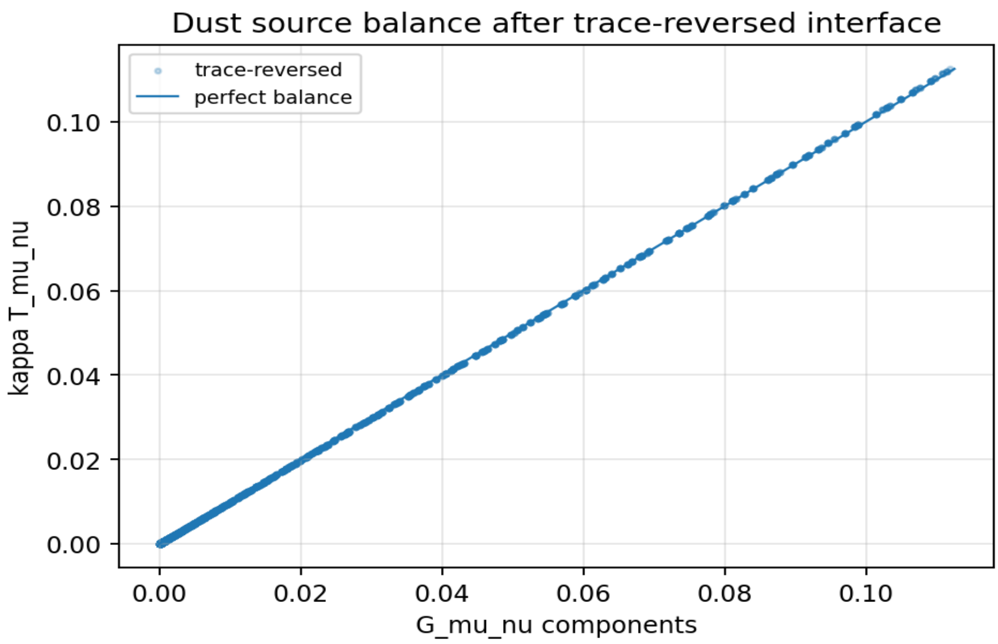



| Case | Residual | Correlation | Euler max | Gauge max | Lorentz sig. |
|---|---|---|---|---|---|
| Dust, trace-reversed | 5.534844e-03 | 0.999985 | 8.046e-14 | 0.000e+00 | 4761/4761 |
| Dust, direct h=hbar | 1.000000e+00 | -0.031486 | 8.046e-14 | 0.000e+00 | 4761/4761 |
| Anisotropic, trace-reversed | 3.138810e-01 | 0.946899 | 8.046e-14 | 3.732e-03 | 4761/4761 |
| Anisotropic, direct h=hbar | 4.614411e-01 | 0.881266 | 8.046e-14 | 3.732e-03 | 4761/4761 |

## 6. Curvature balance for the selected interface

With the trace-reversed interface, the generated metric is passed through the same curvature pipeline: Christoffel symbols, Ricci tensor, Ricci scalar, and Einstein-tensor candidate.






| Balance diagnostic | Value |
|---|---|
| Dust trace-reversed fitted kappa | 1.125261e-01 |
| Dust trace-reversed full tensor residual | 5.534844e-03 |
| Dust trace-reversed correlation | 0.999985 |
| Dust direct-interface residual | 1.000000e+00 |
| Dust direct-interface correlation | -0.031486 |

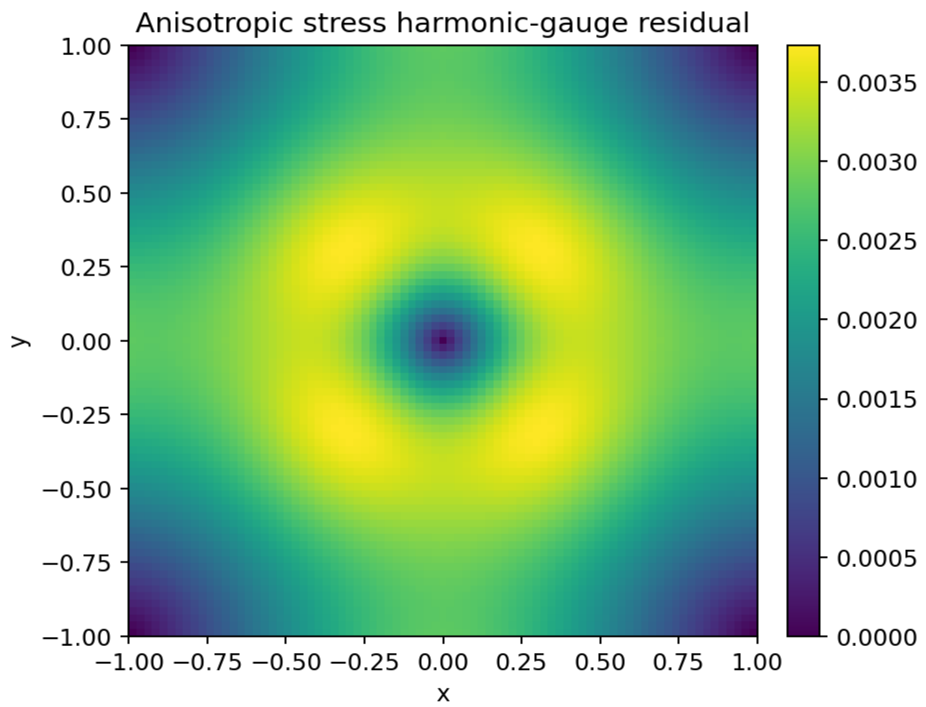
## 7. Component-by-component check

| Component | Residual | Correlation | Max \|G\| | Max \|kT\| |
|---|---|---|---|---|
| tt | 5.534844e-03 | 0.999990 | 1.119311e-01 | 1.125261e-01 |
| tx | 0.000000e+00 | 1.000000 | 0.000000e+00 | 0.000000e+00 |
| ty | 0.000000e+00 | 1.000000 | 0.000000e+00 | 0.000000e+00 |
| xx | 4.354061e-02 | 1.000000 | 6.938894e-18 | 0.000000e+00 |
| xy | 1.000000e+00 | 0.000000 | 5.342850e-08 | 0.000000e+00 |
| yy | 4.356625e-02 | 1.000000 | 6.938894e-18 | 0.000000e+00 |

## 8. Anisotropic stress probe

A second source with spatial pressure and shear was tested to check whether arbitrary tensor sources automatically work. They do not. The tensor action still solves the component equations, but the harmonic-gauge residual and full balance become worse.

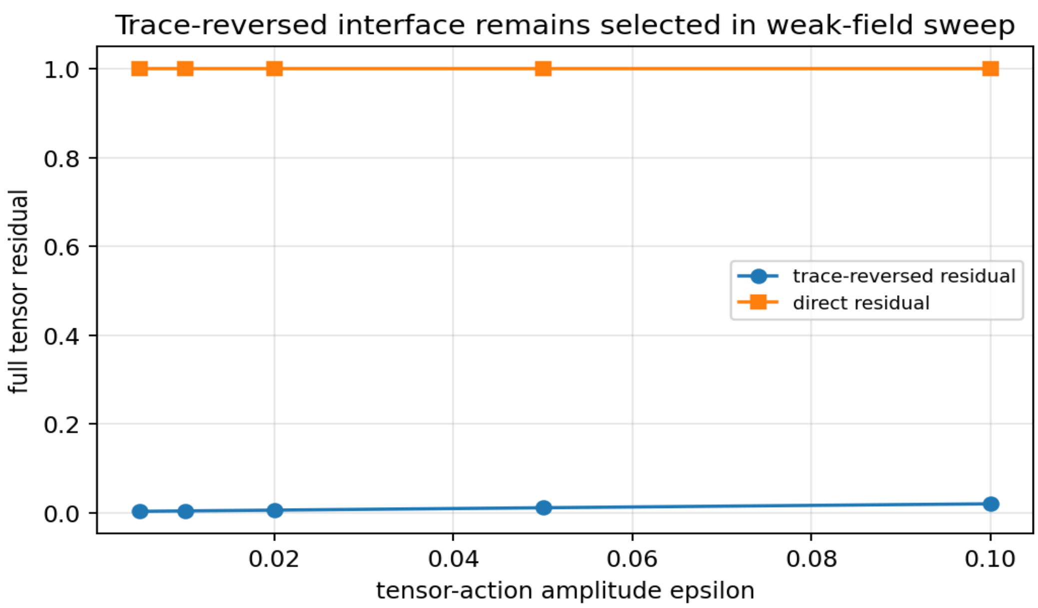
```
Anisotropic probe:
T_tt = S
T_xx = 0.15 S
T_yy = 0.15 S
T_xy = 0.05 (xy/sigma^2) S
```




Interpretation: arbitrary tensor stress cannot simply be inserted. A valid source must satisfy gauge/conservation constraints and must be coupled through a more complete spin-2 action.

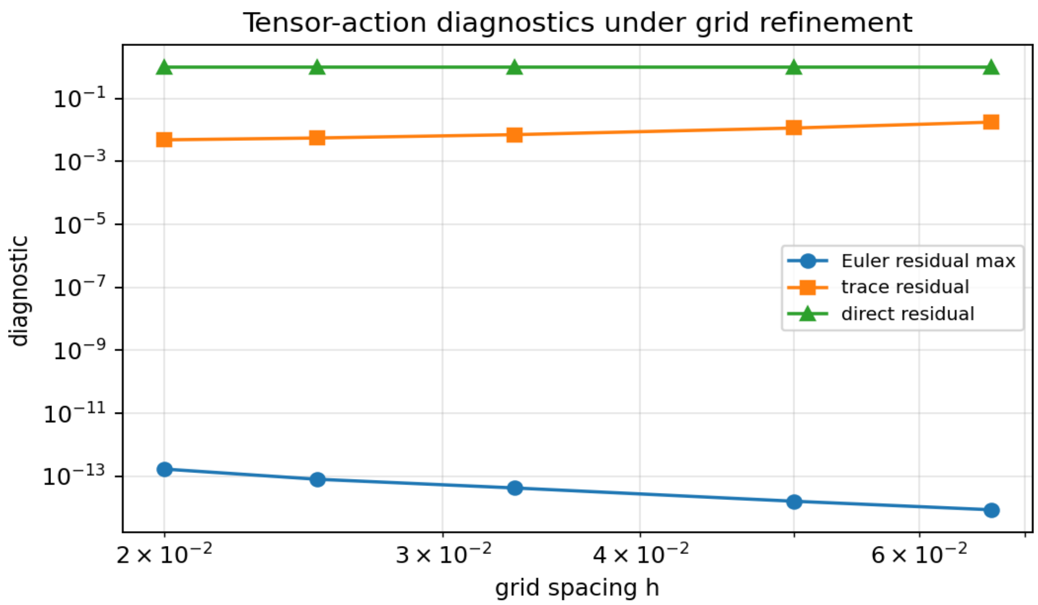
## 9. Weak-field sweep

The dust-source case was tested under different perturbation amplitudes. The trace-reversed interface remains the selected interface across the weak-field sweep.




| epsilon | trace residual | trace corr. | direct residual | direct corr. | Lorentz sig. |
|---|---|---|---|---|---|
| 0.005 | 2.881502e-03 | 0.999996 | 1.000000e+00 | -0.031531 | 4761/4761 |
| 0.010 | 3.759728e-03 | 0.999993 | 1.000000e+00 | -0.031516 | 4761/4761 |
| 0.020 | 5.534844e-03 | 0.999985 | 1.000000e+00 | -0.031486 | 4761/4761 |
| 0.050 | 1.088528e-02 | 0.999942 | 1.000000e+00 | -0.031397 | 4761/4761 |
| 0.100 | 1.976698e-02 | 0.999808 | 1.000000e+00 | -0.031247 | 4761/4761 |

## 10. Grid refinement




| N | h | Euler max | trace residual | direct residual | Lorentz sig. |
|---|---|---|---|---|---|
| 31 | 0.06667 | 8.632e-15 | 1.763935e-02 | 1.000000e+00 | 529/529 |
| 41 | 0.05000 | 1.613e-14 | 1.148196e-02 | 1.000000e+00 | 1089/1089 |
| 61 | 0.03333 | 4.249e-14 | 7.078366e-03 | 1.000000e+00 | 2809/2809 |
| 81 | 0.02500 | 8.046e-14 | 5.545654e-03 | 1.000000e+00 | 5041/5041 |
| 101 | 0.02000 | 1.696e-13 | 4.834121e-03 | 1.000000e+00 | 7569/7569 |

## 11. Core numerical record

| Diagnostic | Value |
|---|---|
| Grid | 81 x 81 |
| Interior points | 4761 |
| Lorentzian signatures | 4761/4761 |
| Source center | 1.000000e+00 |
| hbar_tt center | 2.000000e-02 |
| g_tt center, trace-reversed | -2.890000e+00 |
| g_tt center, direct | -2.870000e+00 |
| G_tt center, trace-reversed | 1.119311e-01 |
| G_tt center, direct | 7.037457e-07 |
| Ricci scalar center, trace-reversed | 7.746094e-02 |

## 12. Interpretation inside the V4.2 framework

Toy Model 4I supports this limited ladder:

```
tensorial relational action A[hbar]
      -> variation with respect to hbar_mu_nu
      -> component-wise tensor Poisson equations
      -> trace-reversed interface selected by balance test
      -> generated metric
      -> weak-field curvature balance
```

It does not yet support this stronger ladder:

```
pure graph principles alone
      -> unique trace-reversal
      -> harmonic gauge
      -> full spin-2 Fierz-Pauli action
      -> nonlinear Einstein-Hilbert action
      -> physical gravity
```

## 13. What Toy Model 4I achieves

• It upgrades the response variable from scalar Phi to tensor hbar_mu_nu.

- It derives component-wise tensor Poisson response from a tensorial action.
- It shows the trace-reversed interface is strongly selected by weak-field balance.
- It shows the direct h=hbar interface fails the same balance test.
- It identifies that arbitrary anisotropic tensor sources require gauge/conservation structure.
- It narrows the next missing object to a gauge-constrained spin-2 relational action.

## 14. What Toy Model 4I does not achieve

- It does not derive trace reversal uniquely from first principles.
- It does not derive the harmonic gauge constraint.
- It does not derive the Fierz-Pauli spin-2 action.
- It does not derive the nonlinear Einstein-Hilbert action.
- It does not derive physical stress-energy or Newton's constant.
- It does not solve gravity, dark matter, or dark energy.
- It does not prove physical 3T.

## 15. Close-out statement for 4I

**Toy Model 4I is a meaningful partial derivation. The scalar response of 4H was successfully upgraded to a tensorial response: a native tensor action varied with respect to hbar_mu_nu produces component-wise tensor Poisson equations. The trace-reversed metric interface is not merely cosmetic; it is strongly selected because it yields weak-field balance, while the direct h=hbar interface fails. However, this is still selection by consistency, not a full first-principles derivation. The missing step is now clear: a gauge-constrained spin-2 relational action that makes trace reversal and harmonic gauge unavoidable.**

## 16. Recommended next chapter

The next logical chapter is Toy Model 4J: Gauge-Constrained Spin-2 Relational Action.

1. Add a gauge/harmonic constraint term to the tensorial action.
2. Use the Fierz-Pauli structure as the benchmark, not as an assumed final truth.
3. Vary the constrained action and test whether trace reversal and gauge conditions are jointly selected.
4. Test conserved and anisotropic stress sources again.
5. Only after this should nonlinear gravitational action be discussed.

## 17. Rebind prompt for continuation

```
We are resuming after Toy Model 4I.

4H:
Derived scalar graph Poisson response from action:
A[Phi] = integral [1/2 |grad Phi|^2 - S Phi] dA
=> nabla^2 Phi = -S

4I:
Upgraded response variable to tensor hbar_mu_nu.

Tensor action:
A[hbar] = integral [1/2 sum |grad hbar_mu_nu|^2
                    - sum T_mu_nu hbar_mu_nu] dA

Variation:
delta A / delta hbar_mu_nu = 0
=> nabla^2 hbar_mu_nu = -T_mu_nu

Metric interfaces tested:
1. direct: h_mu_nu = hbar_mu_nu
2. trace-reversed:
   h_mu_nu = hbar_mu_nu - eta_mu_nu hbar_trace/(D-2)

Result:
Trace-reversed interface is strongly selected by weak-field balance.
Direct h=hbar interface fails.
An arbitrary anisotropic tensor source solves the tensor action but worsens gauge/balance,
showing gauge/conservation structure is missing.

Critical guardrail:
4I does not uniquely derive trace reversal, harmonic gauge, Fierz-Pauli action,
```

Einstein-Hilbert action, physical gravity, dark matter, dark energy, or physical 3T.

Next recommended chapter:
Toy Model 4J - Gauge-Constrained Spin-2 Relational Action.